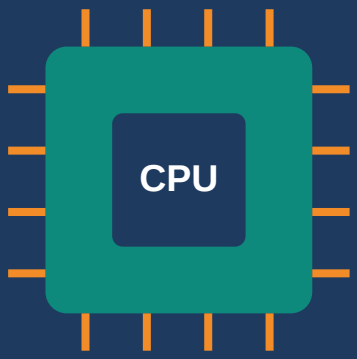
AR-RADAII IMPERIAL ACADEMY

DIGITAL TECHNOLOGY

JSS 2 • First Term

Illustrated Revision Lesson Notes



Weeks covered: 1, 2, 3, 4, 6, 8, 9, 10

Includes: illustrations, board summaries, quick checks, revision quiz + answers

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Term Overview

This is a **revision-focused** set of notes for JSS 2. Each week has objectives, key notes, a board illustration, a board summary and quick-check questions. A revision quiz with answers is at the end.

Week	Topic	Key points
1	Review of JSS1 Work	Hardware, software and online safety recap
2	Basic Computer Communication Networks	Meaning, need, LAN / MAN / WAN
3	Components of a Computer Network	Servers, clients, switches, routers, hubs; cables and wireless
4	Network Topologies	Bus, star, ring; advantages and disadvantages
5	Mid-Term Test	Weeks 1-4
6	Internet and Web Services	Search engines, browsers, email setup and etiquette
7	Mid-Term Break	Holiday
8	Cloud Computing	Cloud storage, Google Drive, Dropbox, advantages and safety
9	Cybersecurity Awareness	Meaning, cyberattacks, safe online habits
10	Data Protection	Meaning, importance, legal and ethical responsibilities
11-12	Revision and Examination	Revision quiz at the back of this document

How to use in class

Read the objectives, show the illustration on the board, go through the key notes, then finish with the Quick Check. Answers are on the last page.

WEEK 1

Review of JSS1 Work

Hardware | Software | Online safety

Objectives

- Recall computer hardware and software.
- Recall basic online safety rules.

Fig 1.1: JSS1 recap

Topic	Quick recap
Hardware	Physical parts. **Input:** keyboard, mouse, scanner. **Output:** monitor, printer, speaker. **Storage:** RAM, ROM (primary); hard disk, flash drive (secondary).
Storage units	8 bits = 1 byte; 1 KB = 1,024 bytes; then MB, GB, TB (each 1,024 times the last).
Software	Programs. **System software:** operating system (Windows, Android). **Application software:** MS Word, browsers, games.
Booting	Starting the computer and loading the OS: power on, POST, BIOS, OS loads, desktop. Always shut down properly.
Online safety	Use strong passwords, do not share personal details, do not open unknown links, report bullying, log out on shared computers.

Strong password example: use 8 or more characters mixing capital letters, small letters, numbers and symbols. Do not use your name or birthday.

Board Summary
Hardware is what you can touch; software is the programs. Storage is measured in bytes. Stay safe online with strong passwords and caution.

Quick Check (ask the class)

1. Differentiate between hardware and software.
2. Give two examples each of input and output devices.
3. State two online safety rules.

WEEK 2

Basic Computer Communication Networks

Meaning and need for networks | LAN, MAN, WAN

Objectives

- Define a computer network.
- State the need for networks.
- Differentiate between LAN, MAN and WAN.

1. Meaning of a Network

A **computer network** is two or more computers and devices connected together so they can **share data, resources and communicate**.

2. Need for Networks (Advantages)

- **Resource sharing:** many users share one printer, scanner or Internet connection.
- **Fast communication:** email, chat, video calls.
- **File sharing** and central storage/backup of data.
- Saves cost and allows easy management of computers.

3. Types of Networks

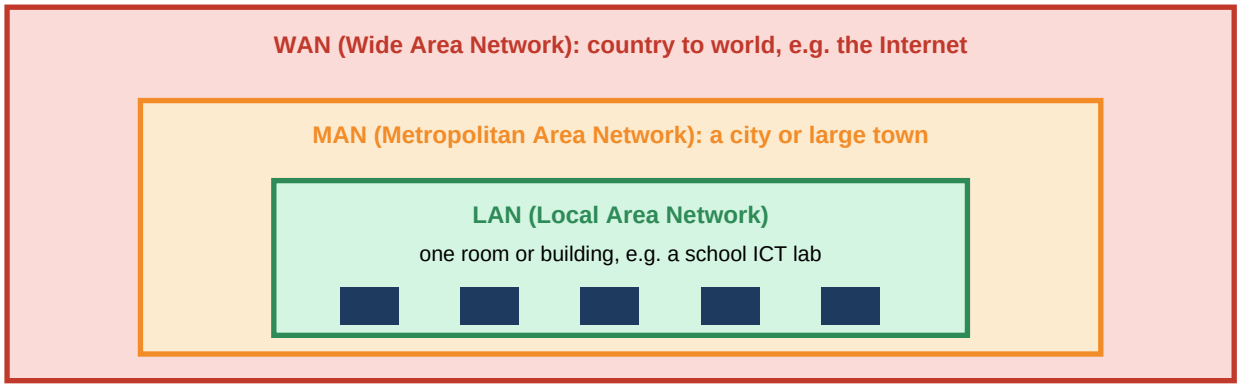

Fig 2.1: LAN inside MAN inside WAN

Type	Full meaning	Coverage	Example
LAN	Local Area Network	Small: a room, building or school	School computer lab, home Wi-Fi
MAN	Metropolitan Area Network	A city or large town	Cable TV network in a city, bank branches in a city
WAN	Wide Area Network	Very large: countries and the world	The Internet

Board Summary
A network connects computers to share resources and communicate. LAN is small, MAN covers a city, WAN covers the world (the Internet).

Quick Check (ask the class)

1. Define a computer network.
2. Give three advantages of networking.
3. What does LAN stand for?

WEEK 3

Components of a Computer Network

Servers, clients, switches, routers, hubs | Network media

Objectives

- Identify the main network components and their functions.
- Differentiate between wired and wireless media.

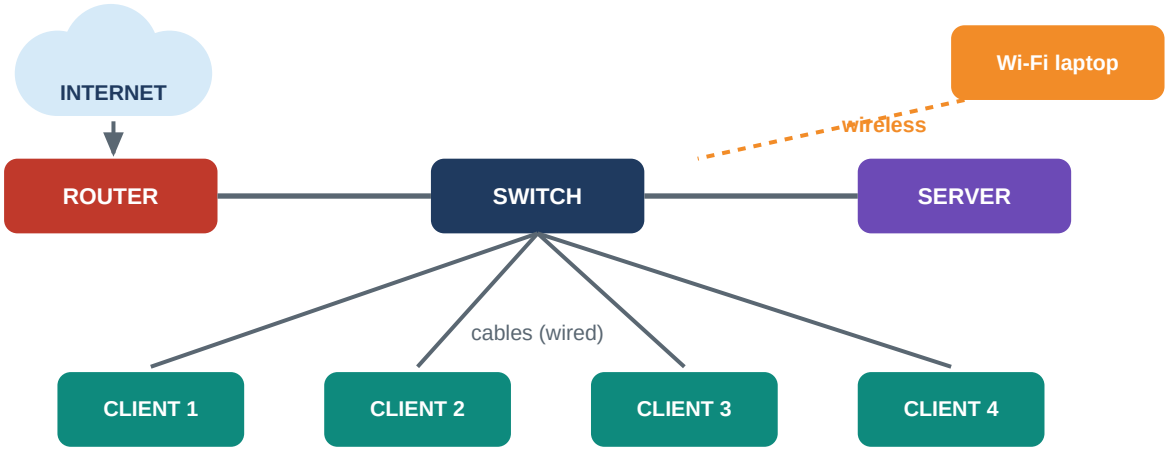

Fig 3.1: A simple network with its components

Component	Function
Server	Powerful computer that provides services, files and resources to other computers
Client	Computer or device that uses the services of a server (workstation)
Hub	Basic device that connects computers and sends data to **all** of them
Switch	Smarter than a hub: sends data only to the **correct** computer
Router	Connects different networks, e.g. a LAN to the Internet; directs data

Network Media

Wired (cables)	Wireless
Twisted pair (Ethernet/UTP) cable, coaxial cable, fibre optic cable (very fast, uses light)	Wi-Fi, Bluetooth, radio waves, infrared, satellite

Board Summary
Servers provide, clients use. Hubs broadcast, switches send to the right device, routers link networks. Media can be cables or wireless.

Quick Check (ask the class)

1. Differentiate between a server and a client.
2. What does a router do?
3. Give two examples of wireless media.

WEEK 4

Network Topologies

Bus | Star | Ring | Advantages and disadvantages

Objectives

- Define network topology.
- Describe bus, star and ring topologies.
- State advantages and disadvantages of each.

Network topology is the way computers and devices are arranged and connected in a network.

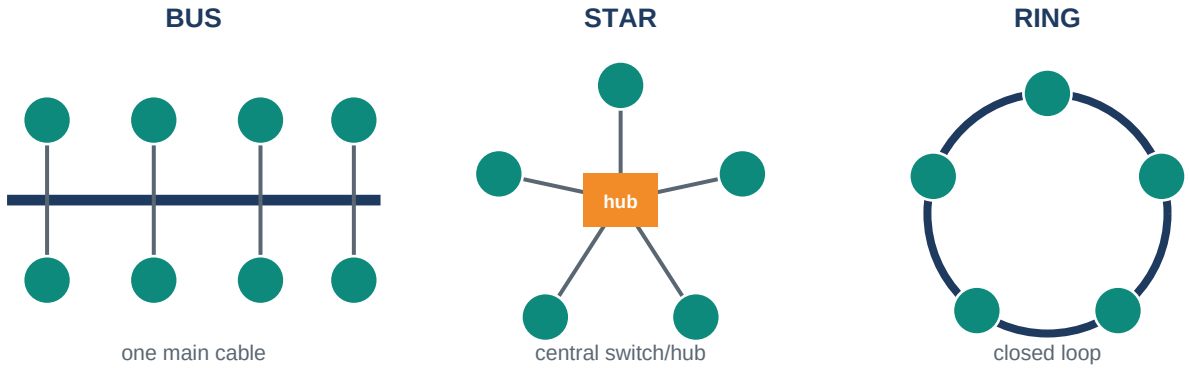

Fig 4.1: Bus, star and ring topologies

Topology	Advantages	Disadvantages
Bus: all devices on one main cable	Cheap, easy to set up, needs little cable	If the main cable fails the whole network fails; slow with many users
Star: all devices connect to a central hub/switch	Easy to add devices; one faulty cable does not stop others; easy to find faults	If the central hub fails the whole network fails; needs more cable
Ring: devices form a closed loop	Orderly data flow; works well for few users	One broken device or cable can stop the network; hard to add devices

Board Summary
Topology = layout of a network. Bus uses one cable, star uses a central switch (most common today), ring forms a loop.

Quick Check (ask the class)

1. Define network topology.
2. Name one advantage and one disadvantage of star topology.
3. Which topology uses a single main cable?

WEEK 5

Mid-Term Test

Assessment of Weeks 1-4

Use the Quick Check questions from Weeks 1-4 and Section A of the Revision Quiz as a short test. Suggested format: 10 objective questions + 3 short-answer questions (20 to 30 minutes).

Reminder (Week 7)
Week 7 is mid-term break. There are no classes. Notes resume with Week 8.

WEEK 6

Internet and Web Services

Search engines and browsers | Email setup and etiquette

Objectives

- Differentiate between the Internet and the World Wide Web.
- Differentiate between a browser and a search engine.
- Set up an email and follow email etiquette.

The **Internet** is the worldwide network of networks. The **World Wide Web (WWW)** is the collection of web pages you access on the Internet.

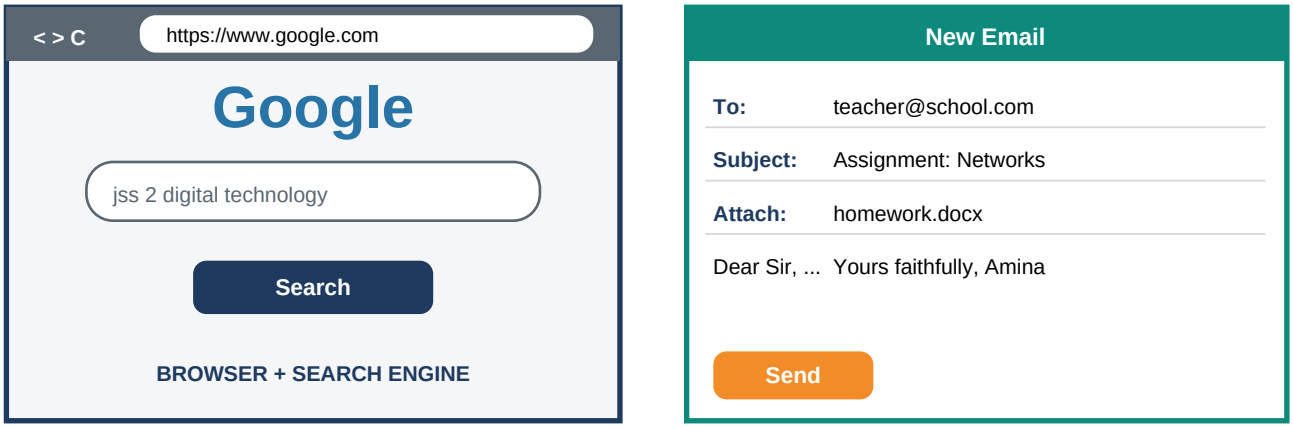

Fig 6.1: Browser with search engine; composing an email

Item	Meaning	Examples
Browser	Software used to open and view web pages	Chrome, Firefox, Edge, Safari
Search engine	Website that finds information on the web from keywords	Google, Bing, DuckDuckGo

Email Setup (e.g. Gmail)

Go to the email website, click **Create account**, enter name, choose a username, create a strong password, verify with phone number if asked, and accept the terms. Then **Compose**, fill To, Subject and message, attach files if needed, and click **Send**.

Email Etiquette

- Write a clear **subject**; start with a greeting and end politely.
- Be respectful; do not use ALL CAPITALS (it looks like shouting); check spelling before sending.
- Do not open unknown attachments or reply to spam; keep messages short.

Board Summary
A browser opens web pages; a search engine finds information. Create an email with a strong password and write politely and clearly.

Quick Check (ask the class)

1. Differentiate between a browser and a search engine.
2. State two rules of email etiquette.
3. Give two examples of search engines.

WEEK 8

Cloud Computing

Cloud storage | Google Drive and Dropbox | Advantages and safety

Objectives

- Explain cloud computing and cloud storage.
- Give examples of cloud services.
- State advantages and safety tips.

Cloud computing is using the Internet to store, access and run files and programs on remote servers instead of only on your own device. **Cloud storage** keeps your files online so you can open them from any device with Internet.

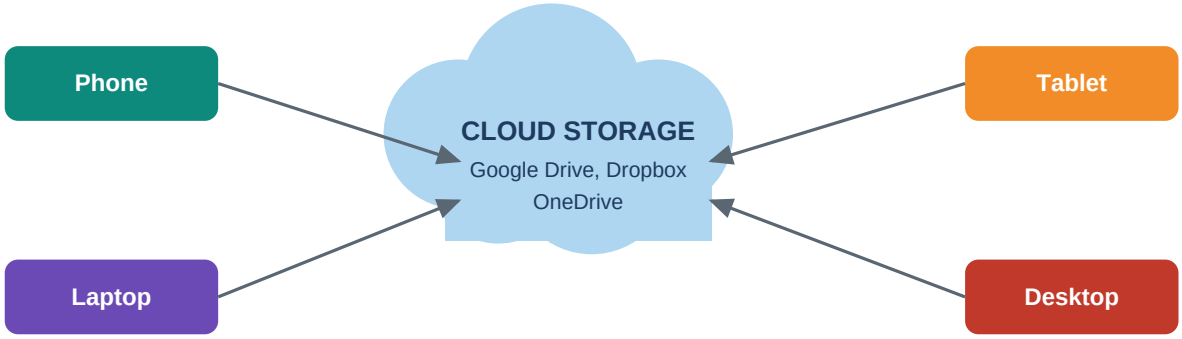

Fig 8.1: Cloud storage shared across devices

Advantages	Disadvantages / Risks
Access files anywhere, on any device; easy sharing and collaboration	Needs Internet access (and data)
Automatic backup, so files are not lost if the device is damaged	Hackers may attack weak accounts; limited free space
Saves space on your device; often free to start	Service provider controls the servers

Examples

Google Drive, **Dropbox**, Microsoft OneDrive, iCloud.

Safety Tips

- Use a **strong password** and turn on **two-factor authentication**.
- Do not share links of private files with everyone; set sharing permissions.
- Log out on shared computers; keep a second backup of important files.

Board Summary
Cloud storage keeps files online so you can reach them from anywhere. Examples: Google Drive, Dropbox. Protect accounts with strong passwords and two-factor authentication.

Quick Check (ask the class)

1. What is cloud storage?
2. Name two cloud storage services.
3. State two safety tips for using the cloud.

WEEK 9

Cybersecurity Awareness

Meaning | Types of cyberattacks | Safe online habits

Objectives

- Define cybersecurity.
- Identify types of cyberattacks.
- Practise safe online habits.

Cybersecurity is the protection of computers, networks, accounts and data from attack, theft and damage.

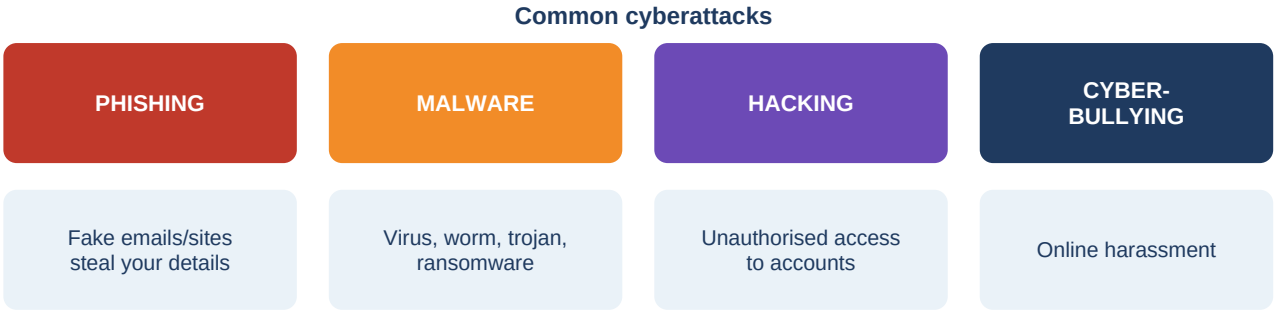

Fig 9.1: Common cyberattacks

Attack	Meaning
Phishing	Fake emails, messages or websites that trick you into giving passwords or bank details
Malware	Harmful software: virus, worm, trojan, spyware, **ransomware** (locks files for money)
Hacking	Gaining access to a system or account without permission
Cyberbullying	Using the Internet or phone to harass, threaten or embarrass someone
Identity theft / scam	Pretending to be someone or offering fake prizes to steal money or information

Safe Online Habits

- Use **strong, different passwords**; turn on two-factor authentication.
- Do not click unknown links or download from unknown sources.
- Install and update **antivirus** and system updates.
- Do not share personal details (address, school, passwords) with strangers.
- Use secure sites (https and the padlock sign); report bullying and tell a trusted adult.

Board Summary
Cybersecurity protects systems and data. Beware of phishing, malware, hacking and cyberbullying. Use strong passwords, antivirus and think before you click.

Quick Check (ask the class)

1. Define cybersecurity.
2. What is phishing?
3. State three safe online habits.

WEEK 10

Data Protection

Meaning and importance | Legal and ethical responsibilities

Objectives

- Define data protection.
- State its importance.
- Explain legal and ethical responsibilities.

Data protection is keeping personal and important information safe from loss, misuse, theft or unauthorised access. **Personal data** includes name, address, phone number, photos, bank details, school records and biometric data like fingerprints.

Life cycle of personal data

Fig 10.1: Responsible handling of data

Importance

- Protects privacy and prevents identity theft and fraud
- Builds trust between people, schools and businesses
- Prevents loss of important data (backup) and legal trouble

Legal and Ethical Responsibilities

Legal (the law)	Ethical (doing what is right)
In Nigeria, the **Nigeria Data Protection Act (2023)** protects personal data and is enforced by the **Nigeria Data Protection Commission (NDPC)**. Organisations must collect data lawfully, store it safely and not misuse it.	Get **consent** before sharing someone's data or photos; collect only what is needed; keep passwords private; be honest; respect other people's privacy; delete data you no longer need.

Board Summary
Data protection keeps personal information safe. The law (Nigeria Data Protection Act) and good ethics require consent, safe storage and no misuse.

Quick Check (ask the class)

1. Define data protection.
2. Give three examples of personal data.
3. State one legal and one ethical responsibility.

WEEK 11

Revision Quiz

Review of term's topics

Section A: Objective (choose the correct answer)

1. The keyboard is an _____ device.
2. A network that covers a school computer lab is a _____.
3. The Internet is an example of a _____.
4. A computer that provides services to others is a _____.
5. The device that connects a LAN to the Internet is a _____.
6. Which sends data to the correct computer only? Hub / Switch
7. Fibre optic cable transmits data using _____.
8. The topology where all devices connect to a central device is _____.
9. The topology that uses one main cable is _____.
10. Chrome and Firefox are examples of _____.
11. Google and Bing are examples of _____.
12. Google Drive and Dropbox are examples of _____ storage.
13. A fake email that steals your details is _____.
14. Ransomware is a type of _____.
15. The body that enforces data protection in Nigeria is the _____.

Section B: Short answer

16. Define a computer network and state two advantages.
17. Differentiate between LAN, MAN and WAN.
18. State the functions of a hub, a switch and a router.
19. Give one advantage and one disadvantage of star topology.
20. State three rules of email etiquette.
21. Mention three advantages of cloud storage.
22. List three safe online habits.
23. Define data protection and give two examples of personal data.

Answer Key

No.	Answer
1	input
2	LAN
3	WAN
4	server
5	router
6	Switch
7	light
8	star
9	bus
10	browsers
11	search engines
12	cloud
13	phishing
14	malware
15	NDPC (Nigeria Data Protection Commission)
16	Two or more connected computers that share data and resources. Advantages: resource sharing, fast communication, file sharing.
17	LAN: small area (building); MAN: a city; WAN: very large, countries/world.
18	Hub sends data to all; switch sends to the correct device; router connects networks (e.g. to the Internet).
19	Advantage: a faulty cable does not stop others. Disadvantage: if the central device fails, the network fails.
20	Clear subject, polite greeting, no all capitals, check spelling, do not open unknown attachments.
21	Access anywhere, automatic backup, easy sharing, saves device space.
22	Strong passwords, do not click unknown links, use antivirus, do not share personal details.
23	Keeping personal information safe from misuse or loss; e.g. name, phone number, address, photos.

Quick Check answers (suggested)

- **Wk1:** Hardware is physical, software is programs; keyboard/mouse and monitor/printer; strong passwords, do not click unknown links.
- **Wk2:** Connected computers sharing resources; resource sharing, fast communication, file sharing; Local Area Network.
- **Wk3:** Server provides services, client uses them; router connects networks; Wi-Fi, Bluetooth.
- **Wk4:** Layout of a network; star: easy to add devices / central failure; bus.
- **Wk6:** Browser opens web pages, search engine finds information; clear subject, politeness; Google, Bing.
- **Wk8:** Storing files online; Google Drive, Dropbox; strong password, two-factor authentication.
- **Wk9:** Protecting systems and data from attack; tricking people with fake messages; antivirus, strong passwords, no unknown links.
- **Wk10:** Keeping personal information safe; name, phone, address, photos; NDPA 2023 / getting consent.